# Matevz Matjasec

**AI RESEARCHER · MAX PLANCK INSTITUTE FOR INTELLIGENT SYSTEMS**

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## RESEARCH PROFILE

I investigate the principles of intelligence through mechanistic analysis of optimization, learning objectives and architectural inductive biases, aiming to develop adaptive, robust and generalizable systems. My work spans optimizer design, meta- and in-context learning, diffusion language models, sequence factorization, linear models and long-context modelling, with applications in genomic language modelling and single-cell transcriptomics.

## WORKSHOP PAPERS

**Conditional Inference Mismatch in Structured Diffusion Language Models**.
First author · ICML 2026, SPIGM Workshop (Structured Probabilistic Inference & Generative Modeling).

**Optimizers Select Algorithms: Update Geometry Controls Memory Strategy in Associative Recall**.
First author · NeurIPS 2026, OPT Workshop (Optimization for Machine Learning).

## RESEARCH IN PROGRESS

- **Diffusion, objectives & ICL —** Investigating objective factorization, causal inductive biases and conditional inference in discrete diffusion language models.
- **Single-cell representation learning —** Exploring transcriptome-wide generative modelling using masked diffusion and linear attention states.
- **Associative recall —** Comparing contextual and memorized recall across architectures and optimization methods.

## CURRENT RESEARCH

**AI Researcher | Max Planck Institute for Intelligent Systems** **2026–present**

**Advisor:** Antonio Orvieto

- **Optimization and learning dynamics:** investigating how update geometry, inductive biases and model architecture shape learned algorithms, generalization and memory, using mechanistic probes and interpretability analyses.
- **Diffusion and adaptive learning:** investigating new training objectives and factorization in discrete diffusion; exploring iterative refinement, looping, semantic diffusion, post-training methods and test-time training.
- **Machine learning for biology:** genomic language models and sequence-to-function transfer; generative representations of single-cell transcriptomes and perturbation responses.
- **Additional theoretical directions:** GNN oversmoothing and its connections to reservoir computing; long-context modelling.

## EDUCATION

**MPhil, Advanced Computer Science | University of Cambridge** 2022–2023

Research thesis (supervisor: Andreas Vlachos): shortcut learning in pretrained language models, analysed through optimization dynamics and loss landscapes.

**BSc, Physics — First Class Honours | King's College London** 2019–2022

Research: protein-family classification with ProtBERT; additional work in computational physics and machine learning. Top 5% of cohort in first year.

## SELECTED PRESENTATIONS & ACADEMIC SERVICE

- **11 Dec 2026 · NeurIPS OPT Workshop, Sydney (upcoming).** Presenting accepted first-author work on optimizer update geometry and memory strategies in associative recall.
- **21 Aug 2026 · MPI-IS FridayTalks, Tübingen.** Presented research on Causal Inductive Biases for In-Context Learning in Diffusion Language Models.
- **9 Aug 2026 · Petnica Summer Institute (ML-SCI), Serbia.** Gave a talk on genomic language models, sequence-to-function modelling and training objectives to undergraduate and graduate students; organized a student workshop project.
- **2026 · FridayTalks, Tübingen.** Organizing committee; coordinating talks and cross-group research exchanges in the Tübingen AI community.
- **May 2026 · Cold Spring Harbor Laboratory.** Attended the 90th Symposium on Quantitative Biology, "AI in Biology."

## PRIOR RESEARCH & PROFESSIONAL EXPERIENCE

**ML & AI Advisor | Alpdev** Oct 2025–present

Advising on applied machine learning, advanced analytics and AI-driven software systems.

**Research Assistant | Jožef Stefan Institute** Dec 2021–Apr 2022

EU Humane AI Net project: analysis of a multilingual news corpus and language-model-based methods for event clustering.

**Selected entrepreneurial work** 2023–2025

Faraday Synergy: intelligent coordination of distributed household/EV energy storage; OBED.ai: multilingual NLP-based voice agents.

## AWARDS & ACADEMIC DISTINCTIONS

- **Ad Futura Scholarship for Study Abroad** (2022) — national Slovenian scholarship for international study.
- **Rotary Club Ljubljana Scholarship** (2022) — awarded for academic excellence in the natural sciences.
- **Ljubljana Scholarship for International Undergraduate Students** (2021).
- **Granted patent** (2022) — weight-adjusted running bib; additional 2021 patent applications in traffic optimization and satellite power transmission.
- **International Theoretical Physics Olympiad** — represented King's College London in the international competition.
- **Princeton Physics of Life Summer School** (2020) — interdisciplinary programme covering biophysics, neuroscience, machine learning and complex systems.

## TECHNICAL SKILLS

Python (PyTorch, NumPy), R, LaTeX · Deep learning, optimization, transformers, masked diffusion, linear attention, GNNs, biological sequence modeling